



Martina Mozzanica

PhD Candidate · Machine Learning for High Energy Physics

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Group page, University of Hamburg

■ PROFILE

PhD candidate in generative machine learning at the University of Hamburg, building fast, geometry-independent calorimeter simulations with point-cloud generative models. Earlier work at CERN and Milano-Bicocca developed graph neural networks for the LHCb trigger. Experienced across the HEP software chain, from Geant4 and Key4hep to PyTorch training on GPU clusters.

■ RESEARCH EXPERIENCE

Research Stay · CERN, Geneva

May 2026 – Jul 2026

- Built an end-to-end pipeline (Geant4 → Key4hep → step2point → CaloClouds3) to study how the choice of point-cloud representation affects a generative shower model for the ILD calorimeter.

PhD Candidate · University of Hamburg

Sep 2023 – present

- Generative machine learning for high energy physics in the group of Prof. Gregor Kasieczka.
- Co-developed point-cloud generative models (transformers, normalizing flows) for fast, highly granular calorimeter simulation across particles and detector geometries.

Research Fellow · Università degli Studi di Milano-Bicocca

Apr 2023 – Jul 2023

- Continued master's thesis work: graph neural networks for the DFEI project at LHCb.

Research Stay · CERN, Geneva · Deep Full Event Interpretation (DFEI) project, LHCb

Jun 2023 – Jul 2023

Research Stay · CERN, Geneva · Master's thesis with the DFEI group

Oct 2022 – Nov 2022

■ EDUCATION

PhD in Physics (ongoing) · University of Hamburg

Sep 2023 – present

M.Sc. in Physics, High Energy Physics · Università degli Studi di Milano-Bicocca

Oct 2020 – Mar 2023

Thesis: graph neural networks for the LHCb trigger, with the DFEI group at CERN.

Erasmus+ Exchange, Engineering Physics · Technical University of Munich

Oct 2021 – Mar 2022

B.Sc. in Physics, Applied Physics · Università degli Studi dell'Insubria

2017 – 2020

Diploma, Liceo Scientifico · Liceo Scientifico Statale "G. Ferraris"

2012 – 2017

■ PUBLICATIONS

- CaloHadronic: A Diffusion Model for the Generation of Hadronic Showers
- Point-Cloud Generative Models for Fast Calorimeter Simulation Across Particles and Geometries
- SPADE: Split-and-Delay Embeddings for Autoregressive High-Granularity Calorimeter Simulation
- OmniJet- α C: Learning Point Cloud Calorimeter Simulations Using Generative Transformers
- CaloClouds3: Ultra-Fast Geometry-Independent Highly-Granular Calorimeter Simulation
- Jevons' Paradox and Fast Generative Simulation for HEP: Why Realistic Benchmarking Is Essential
- GNN for Deep Full Event Interpretation and Hierarchical Reconstruction of Heavy-Hadron Decays in Proton-Proton Collisions

■ SKILLS

Machine learning

Generative models, diffusion models, normalizing flows, VAEs, model distillation, transformers, graph neural networks, point-cloud models, HDBSCAN, large-scale data analysis

Programming

Python, C++, PyTorch, TensorFlow

HEP software

Geant4, Key4hep, ROOT

■ OTHER EXPERIENCE

- Substitute Teacher**, ISIS Valceresio · May 2021
- Referee**, Centro Sportivo Italiano · 2016 – 2017
- Certified Artistic Gymnastics Judge**